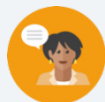


3.7 Absolute Value

Lesson Introduction

 Benchmark	MA.6.NSO.1.3 Given a mathematical or real-world context, interpret the absolute value of a number as the distance from zero on a number line. Find the absolute value of rational numbers.		 Learning Targets	- I can find the absolute value of a rational number. - I can find the absolute value of a value and interpret it within the context.
 Benchmark Coherence	Building On N/A		Working Toward MA.7.NSO.2.1 Solve mathematical problems using multi-step order of operations with rational numbers including grouping symbols, whole-number exponents and absolute value.	
 Essential Questions	- How can you determine any number's distance from zero on a number line? - What is the relationship between the absolute value of opposites? - What represents zero in the real-world situation? - How do you determine and interpret the absolute value in real-world situations? - How do you determine the absolute value of a number?			
 Lesson Narrative	Students formally define and make meaning of absolute value in this lesson introducing absolute value. Students practice identifying absolute value and plotting it on a number line, then extend their working definitions of absolute value to include sense-making in real-world contexts			
 Component Summary	3.7.1 Warm-Up (~5 min.)	Justify which stack of coins they would rather have (<i>SE p. 111</i>)	3.7.4 Guided Practice (10 min.)	Guided practice determining the absolute value using a number line (<i>SE p. 113</i>)
	3.7.2 Exploration (5-10 min.)	Exploration of distance to zero and opposites using a number line (<i>SE p. 112</i>)	3.7.5 Collaboration (10 min.)	Determine and interpret absolute value and opposites in real-world situations (<i>SE p. 114</i>)
	3.7.3 Guided Instruction (5 min.)	Learn the meaning of absolute value (<i>SE p. 113</i>)	3.7.6 Your Turn! (5-10 min.)	Additional practice determining the absolute value using a number line and interpreting absolute values and opposites in real-world situations (<i>SE p. 116</i>)
	3.7.7 Wrap-Up (~5 min.)	Demonstrate understanding of lesson learning goals (<i>SE p. 116</i>)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> - Absolute value - Speed <u>Additional Terms:</u> N/A		Number line templates	
			Additional Preparation	
			3.7.2 Exploration prompts students to use a bug cutout in the number line example.	

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may struggle estimating the location of fractions and decimals when plotting rational numbers on the number line. - Students may incorrectly think that the distance of a negative integer from zero is a negative number or that the absolute value of a number is its opposite. - Students may believe larger negative numbers also hold larger value (i.e., -7 is greater than -6).	- Avoid defining absolute value as the positive value, as this does not build conceptual understanding, which will have impacts for their understanding of adding and subtracting integers in Unit 6. Focus instead on students making sense of absolute value as the number's distance from zero. - Students may benefit from having number lines to reference throughout the lesson. - If time is an issue, consider assigning 3.7.6 Your Turn! for homework.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder In 3.7.3 Guided Instruction, ask students what they notice and wonder about the absolute values of opposite numbers. Facilitate a discussion to ensure students understand why the absolute values of opposites are equal.	MA.K12.MTR.4.1: Discussion and Reflection 3.7.4 Guided Practice provides an opportunity for students to discuss student work. Students discuss how it could be possible for more than one student answer to be correct to a single problem. Students will build a stronger understanding of the concepts through this discussion.
 Differentiate Instruction	3.7.2 Exploration provides kinesthetic entry points for students who better make sense of concepts using manipulatives and real-world spatial awareness. Consider differentiated approaches to this activity based on your students. - Utilize the bug cutout and number line for students to move along the line. - Construct a life-size number line with students acting as the bugs.	
Lesson Notes:		

3.7.1 Warm-Up: Would You Rather?

Component Overview: Students determine which stack of coins they would prefer. Then students justify their selection.



3.7.1 Warm-Up: Would You Rather

1. Would you rather have Pile A or Pile B? Justify your reasoning.



Pile A	Pile B
14 pennies	2 pennies
7 nickels	4 nickels
2 dimes	2 dimes
1 quarter	2 quarters

Answers will vary. I would rather have Pile A because Pile A has \$0.94 and Pile B has \$0.92.



Contemplate, then Calculate

Use this instructional routine to position students as thinkers first, thinking about their approach before immediately jumping into the calculations. This routine provides space and opportunity for meta-cognition, or thinking about their thinking, which ultimately contributes toward fluency.

3.7.2 Exploration: Jumping Bugs

Component Overview: Students explore the distance to zero of several integers, including opposites, using a number line. The exploration occurs through the story of a bug that jumps to the left and the right. The bug's home is at zero.



3.7.2 Exploration: Jumping Bugs

A bug is jumping around on the number line. His home is located at 0. Use a cut out of the bug to answer the following questions about the bug's location.



1. The bug starts at 5 and jumps 3 units to the right.
 - a. What is the bug's new location?
8
 - b. How far away from home is the bug?
8 jumps
2.
 - a. If the bug starts at 2 and jumps 5 units to the left, what is the bug's new location?
-3
 - b. How far away from home is the bug?
3 jumps
3. Mr. Rodriguez told his class to begin with the bug at home. Then the bug should jump 4 units. When Micah and Brianna did this, they discovered that their bugs were in two different locations. Discuss with your partner how this may have happened. Write your explanation or draw a picture of what could have happened.

Answers will vary. Micah and Brianna jumped their bugs in opposite directions, landing on -4 and 4.



Try this activity outside to better engage students in making sense of distance between units, direction of units, and more.

3.7.3 Guided Instruction: Absolute Value and Zero

Component Overview: Students complete guided notes as they learn the meaning of absolute value. Consider creating and displaying an anchor chart with instructions, examples, and visual models for students to reference in the subsequent components.



3.7.3 Guided Instruction: Absolute Value and Zero

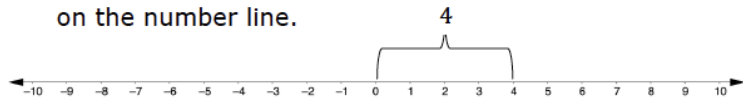
Recall how opposite numbers relate to zero. Complete the explanation below by writing a word in each blank.

1. Pairs of opposite values are the same distance from zero on the number line. They are mirror images of each other, with a line of symmetry through zero.

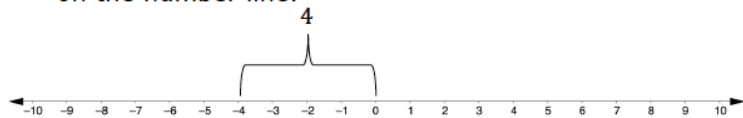


Absolute value is the distance a number is from zero (0) on a number line.

2. The absolute value of 4 is 4 because it is 4 units from 0 on the number line.



3. The absolute value of -4 is 4 because it is 4 units from 0 on the number line.



4. The values 4 and -4 are opposites. Each value is 4 units from zero. A number and its opposite have the same absolute value.



Notice and Wonder

Ask students what they notice and wonder about the absolute values of opposite numbers. Facilitate a discussion to ensure students understand why the absolute values of opposites are equal.

3.7.4 Guided Practice: Absolute Value and Zero

Component Overview: Students practice finding the absolute value of numbers and plotting on a number line.



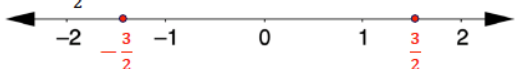
3.7.4 Guided Practice: Absolute Value and Zero

Absolute value is represented using the notation $|a|$. The notation $|a|$ is read, "the absolute value of a ."

1.
 - a. The $|-8| = 8$ because the distance from -8 to 0 is 8 .
 - b. What other number has the same absolute value as -8 ?
 8
 - c. Use the number line to show the other number that has the same absolute value as -8 .



2. On the number line, plot and label all numbers with an absolute value of $\frac{3}{2}$.



3. Hector and his partner, Salih, were given the following two problems to complete.

$$|5.5| = 5.5$$

$$|-3.2| = 3.2$$

3.2 is also correct

Hector and Salih had the same answer for one of the blanks but two different answers for the other. Their teacher told them they both had the correct answers.

- a. Which problem can have more than one correct answer?
The absolute value that is equal to 3.2.
- b. Use the number line to explain how this is possible.
3.2 and -3.2 are the same distance from 0 on the number line, so they both have an absolute value of 3.2.



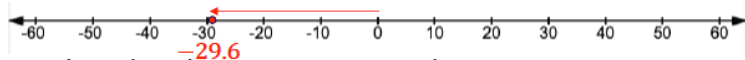
How do you use a number line to determine the distance a number is from zero?

3.7.5 Collaboration: Absolute Value and Opposites in the Real World

Component Overview: Students determine and interpret absolute values in real-world situations. The real-world contexts are related to changes in weight, speed, temperature, and elevation.

3.7.5 Collaboration: Absolute Value and Opposites in the Real World

- A tiger was rescued by The Brevard Zoo in Melbourne, Florida. The zoo veterinarian examined the tiger and determined the animal's weight to be 29.6 pounds below average.
 - What value represents this situation?
 -29.6
 - What is the absolute value of the value that represents the tiger's weight loss?
 $|-29.6| = 29.6$
 - Model the tiger's weight loss on the number line.



- What does 0 represent in this situation?
Answers will vary. Zero represents the change from average weight.
 - If the tiger's weight had been 29.6 pounds above average instead of below, what would be the same about this situation?
Answers will vary. The amount of change would remain the same.
- Simone is doing research on different rides at the Strawberry Festival. She attaches an accelerator to a bumper car to measure the velocity of her car. She recorded the velocities, in miles per hour (mph), in the table.

Speed is the absolute value of velocity.

Time (seconds)	Velocity (mph)	Time (seconds)	Velocity (mph)	Time (seconds)	Velocity (mph)
5	-3.1	20	-2.4	35	-5.2
10	-2.7	25	-6.3	40	3.1
15	4.8	30	2.7	45	-7.8



- When considering the changes in the tiger's weight, what does zero represent?
- How do you model a loss in weight?
- How do you model a gain in average?



Facilitate a conversation about speed and velocity. Provide examples and check for understanding to ensure students understand these terms.

In the context of velocity, the signs represent the direction the bumper car is headed.

3.7.5 Collaboration: Absolute Value and Opposites in the Real World (continued)

- Determine when the bumper car is at the same speed. List the times and the speed of the car at those times.
At 5 seconds and 40 seconds, the car is traveling at 3.1 mph. At 10 seconds and 30 seconds, the car is traveling at 2.7 mph.
 - At what time is the bumper car traveling at the slowest speed?
20 seconds
 - At what time is the bumper car traveling at the fastest speed?
45 seconds
3. Before Bryan goes snow skiing, he asks a friend about the weather. The friend says that the temperature is 5°C away from 0. Select all the temperatures that are possible.
- ☐ -10°C
☒ -5°C
☐ 0°C
☒ 5°C
☐ 10°C
4. A part of the city of New Orleans is 19 feet below sea level. We can use “-19 feet” to describe its elevation, and “|-19| feet” to describe its vertical distance from sea level. In the context of elevation, what do each of the following numbers describe?

Value	Meaning in Context
25 feet	<i>The elevation is 25 feet above sea level.</i>
-25 feet	<i>The vertical distance of the elevation from sea level is 25 feet.</i>
-8 feet	<i>The elevation is 8 feet below sea level.</i>
-8 feet	<i>The vertical distance of the elevation from sea level is 8 feet.</i>



Prompt students to reflect on how Bryan’s friend might be more precise in his mathematical language.

- His friend uses the phrase “away from” – why could this have two possible meanings?
- How could this language be more precise to determine the actual temperature?

3.7.6 Your Turn!

Component Overview: Students engage in additional practice determining the absolute value and plotting those values on a number line.

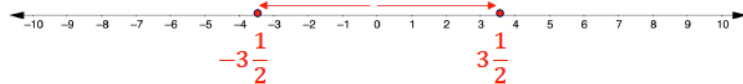
3.7.6 Your Turn!

- Complete each statement and model it on the number line given.

a. $|-7.8| = 7.8$



b. $|\pm 3\frac{1}{2}| = 3\frac{1}{2}$



- The absolute value of the elevation of Enterprise, Florida is 19.7 feet. In relation to sea level, write the two values that could represent the elevation of Enterprise, Florida.

19.7 feet or -19.7 feet



Challenge students to write a real-world example to represent the rational numbers and their absolute values from question 1. Remind students to explain what represents zero in their examples. For example, a student's home is zero, school is 3 miles north (+3), the grocery store is 5 miles south (-5).

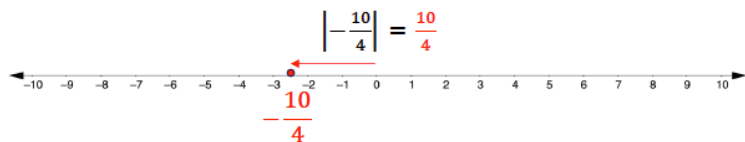
3.7.7 Wrap-Up: Ticket Out the Door

Component Overview: Students demonstrate their mastery of lesson learning targets. Use data from this formative assessment to determine which students need remediation before the next lesson.



3.7.7 Wrap-Up: Ticket Out the Door

- Complete the statement and model it on the number line.



- The number -220 represents a withdrawal in dollars from a checking account, and 220 represents a deposit in dollars to the same checking account. What do the absolute values of these numbers mean in terms of this context?

Answers will vary. The absolute value means that the change in the account balance is \$220 in both situations.



Consider additional information you may want to know from students and add targeted questions to elicit those responses. For example, to elicit responses specific to $\frac{10}{4}$ and $-\frac{10}{4}$ being the same distance from zero, use prompts like: “Write a verbal description for the location of each value on the number line. Compare their location.”